Frontier



atdepth MRV, Inc.

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

atdepth MRV, Inc.

Organization location (we welcome applicants from anywhere in the world)

Boston, MA (US)

Name(s) of primary point(s) of contact for this application

Carlos Muñoz-Royo

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit ocean technology company building operational MRV infrastructure for marine CDR.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Near-field sampling chamber for operational carbonate system monitoring in open-system CDR

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Marine CDR pathways (OAE, RAE, DOR...) rely on carbonate system measurements for quantification and verification. There is consensus that initial deployments are currently being overmeasured through labor-intensive discrete sampling to allow for research activities, uncertainty quantification and methodology improvement. This is desirable in the near term, but it is not a scalable and operational approach. A key bottleneck is the inability of existing off-the-shelf carbonate chemistry sensors to make meaningful measurements in the near-field environments where the most relevant data can be obtained.

Commercially available submersible pCO2 sensors rely on gas equilibration across a membrane to an NDIR detector. Their response times (t63) range from 50 seconds (Pro-Oceanus CO2-Pro CV, the fastest available) to 4 minutes (Turner Designs C-Sense), with full equilibration taking 5 to 15 minutes. In near-field discharge plumes and mixing zones, the signal changes on timescales that are significantly shorter than these response times (O(1)s). Furthermore, the signal can vary by orders of magnitude within seconds due to the turbulent nature of the plumes in the near-field. The result is that we do not know what the sensor readings mean and, as importantly, the reading cannot be tied to a specific water parcel or moment in time.

pH sensors present a different but related set of challenges. ISFET-based sensors (e.g., Sea-Bird SeaFET V2) are the fastest available option, with published response times of a few seconds and a sampling frequency of about 6 seconds per measurement. This is significantly faster than pCO2 sensors, but still insufficient for near-field CDR environments where the signal can change on timescales of seconds. Critically, published response time specifications are characterized for small pH changes under controlled conditions. In a CDR discharge plume where pH may swing by 1 pH unit or more while temperature and salinity are simultaneously fluctuating, the actual dynamic accuracy of the measurement is unknown. The pH reading also depends on a reference electrode (typically a chloride ion-selective electrode) that has its own response dynamics and is sensitive to the same salinity fluctuations present in mixing zones. Solid-state sensors like the ANB Sensors OC300 avoid the reference electrode issue but rely on an internal self-calibration routine that assumes a degree of environmental stability during the calibration sequence, which is not present in near-field conditions. Reagent-based systems (Sunburst SAMI-pH) have measurement cycles of several minutes. In summary, while pH sensors are faster than pCO2 sensors in principle, no available pH instrument has

been demonstrated to make accurate, traceable measurements in the rapidly varying conditions of a CDR discharge plume.

We visualize this mismatch using validated, high-resolution plume simulations (1 s model timestep, 3 m horizontal resolution, validated against analytical solutions and field data). At a representative OAE outfall, the alkalinity signal at 3 meters from the source has a coefficient of variation of 189%, with instances where the signal goes from zero (background water) to a maximum in a matter of tens of seconds. 10m away from the outfall the coefficient of variation still is of 139%, and 89% at 50m. A pH or pCO2 sensor in this environment is not measuring the current state of the water. It is reporting something ambiguous.

The current alternative, discrete sampling with laboratory analysis, is labor-intensive, expensive, and introduces its own uncertainties: samples may take hours to days to analyze, during which carbonate chemistry can evolve, particularly if dissolution or equilibration is not complete at the time of collection.

atdepth has developed a near-field sampling chamber (Figure 1) that solves this challenge by converting a continuously varying signal into a sequence of discrete, stable measurements. The system isolates a water volume at a known time, seals it, and holds it while sensors equilibrate against a stable sample. After the measurement, the chamber is flushed and a new sample is drawn in. Each measurement is tied to a specific water parcel captured at a known moment. The sensor response time ceases to be a limitation because the sample is not changing during measurement. Instead of attempting to develop a fundamentally new sensor, we present existing sensors with a measurement they can actually make in a more reliable and controlled way.

For this project, we propose to build 3 sampling systems using two concepts (a pump-based system we have already bench-tested (Figure 2) and a bladder-based system). The systems will be equipped with a Turner Designs pCO2 sensor and a pH sensor. We will deploy them at two pilot CDR projects that are part of Frontier's portfolio located in very different environments: one project located at a coastal location in the North Atlantic, and the other in a warmer subtropical environment. We have already deployed our operational modeling and monitoring MRV system at the North Atlantic site and are actively working with the project developer. At this site, one system will be mounted on our existing mooring alongside sensors deployed without the chamber, providing direct comparative validation. Deploying across two fundamentally different environments (cold North Atlantic vs. warm subtropical Atlantic) validates the system across the range of conditions and allows testing of

different anti-fouling approaches across two fouling regimes. We are open to discussing the possibility of deploying at a different location.



Figure 1. First prototype of the sampling chamber built by the atdepth team for work-bench testing.

This project sits within our broader operational context as independent MRV provider for mCDR operations. Our team, which includes three key contributors to the Oceananigans ocean modeling framework, has spent the past two years building operational, model-centric MRV infrastructure. Our simulation engine runs full physics-resolving digital twins on a single GPU at very low compute cost. We have already deployed our operational MRV solution at three mCDR projects, and have validated it across more than ten projects in total (some of them non-mCDR related). To serve our projects, we developed the Standard Monitoring Unit (SMU): a cost-effective and purpose-driven monitoring platform to provide the data models need for validation. The near-field sampling chamber idea came from discovering the limitations of existing sensors when exposed to highly variable signals. This project validates the solution we built to address that gap and enable near-field model validation. The SMU is designed for long-term unattended deployment with no specialized personnel and ease of deployment in coastal environments.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

The project addresses three linked research questions:

1. Does the near-field sampling chamber produce reliable, repeatable carbonate system measurements under field conditions, and does it outperform direct sensor exposure?

2. Which sampling concept (pump-based vs. bladder-based) performs better for operational deployment, and under what conditions?

3. What are the dominant failure modes (biofouling, drift, mechanical wear) and how do they vary across deployment environments?

We answer these through a comparative field validation at two active CDR sites, combined with model-measurement integration using our operational digital twin.

We are building and testing two sampling chamber concepts:

1. Pump-based system (already bench-tested): A pump draws water into the sampling chamber and holds it while sensors equilibrate. After measurement the chamber is flushed and a new sample is drawn. This system is mechanically simple and has been bench-tested.
2. Bladder-based system: Captures water volumes passively using a flexible bladder mechanism. This concept offers advantages in sample integrity but introduces greater mechanical complexity. It is designed but not yet prototyped.

Both concepts integrate with our SMU monitoring platform, use the same sensor suite (pH, pCO_2, CTD, dissolved oxygen), and share the same data acquisition and communications architecture.

Deployment plan

Pilot one (2 systems): We will deploy two of the systems at a project site in the North Atlantic coast where we have already deployed monitoring equipment and our modeling system for MRV purposes. One sampling system will be mounted on our existing mooring alongside sensors deployed directly in the water without the chamber. This comparative configuration (same sensors, same location, same conditions, only variable is the chamber) produces a direct quantitative answer to the question of measurement benefit. The second system will be deployed at a different position to quantify the benefits of additional spatial characterization.

Pilot two (1 system): in this case, the deployment will take place at a warmer subtropical environment. This represents a fundamentally different deployment scenario from the North Atlantic one. A single system will be deployed at this location.

We have formal relationships established with both project developers as site access partners. If there are other active deployment sites of interest to Frontier, we are open to considering alternative locations.

Anti-fouling strategy

We consider biofouling management the primary operational risk. The chamber design incorporates multiple anti-fouling provisions that we aim to test in different environments. We will test different combinations of anti-fouling strategies across the two sites to determine which approaches are effective in each environment.

Model-measurement integration

The near-field monitoring data will be directly compared with the physics-based MRV models we already run operationally. Our team is currently operating a full physics-based, end-to-end MRV stack across multiple ocean CDR intervention types simultaneously, with both modeling and monitoring capabilities. As part of the project, we will also assess how to efficiently integrate the near-field measurements with the model.

Decision gates and scope management

Month 6 is the critical decision gate. The key question is whether a sealed chamber produces a measurably better carbonate system reading than direct sensor exposure. The comparative deployment at Pilot one (same mooring, same sensors, chamber vs. no chamber) isolates exactly this variable within the first weeks of deployment. If the chamber shows clear measurement benefit, we proceed to long-term deployment and characterization to assess all the other technical and practical aspects.

The scope is deliberately bounded: 3 systems, 2 sites, 2 concepts, 18 months. We are not developing new sensors, not building a new modeling framework, and not proposing open-ended research.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Biofouling (primary risk, high likelihood, medium impact): Long-term biofouling of the sampling chamber, inlet, and sensors is the main operational risk. We mitigate this through multiple anti-fouling

approaches and by deliberately deploying at two very different fouling environments to characterize which approaches work where. Even if fouling degrades performance over time, the project will produce data on fouling rates and maintenance intervals that is essential for any near-field monitoring deployment in CDR.

Sensor drift (medium likelihood, medium impact): Multi-month deployments may produce sensor drift. The chamber design partially mitigates this by providing controlled, repeatable measurement conditions (consistent sample volume, consistent equilibration time). We will characterize drift rates against discrete reference samples collected during site visits.

Bladder system reliability (medium likelihood, low impact): The bladder-based concept is less mature than the pump-based system. If it proves mechanically unreliable in the field, the pump-based system serves as the validated fallback. Testing both concepts in parallel is a deliberate de-risking strategy.

Deployment logistics (low likelihood, low impact): Permitting, mooring placement, and shipping logistics at both sites are manageable. We already have an existing operational presence at the North Atlantic site (SMU already deployed, ongoing MRV contract) and a confirmed site access partnership for the second pilot. We expect one deployment trip and two inspection visits per site. One of the pilot sites is expected to be operational later in 2026, and delays are always likely. As a mitigation measure, we have planned the deployment for Q2 2027. Additionally, we will identify a third deployment site in case potential delays risk the actual deployment.

Measurement interpretation (low likelihood, medium impact): The comparative design (chamber vs. no-chamber on the same mooring) could produce ambiguous results if the measurement benefit is small relative to other sources of variability. We mitigate this by co-locating the comparison as tightly as possible and by using the model to provide context for interpreting the field data.

Environmental and ecosystem risk (very low): The sampling chamber is a passive measurement device. It does not alter local conditions and poses no environmental risk beyond the physical presence of the mooring and instrument package, which is standard oceanographic practice.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	1. Three validated near-field sampling systems (two concepts: pump-based and bladder-based) deployed and tested at two active CDR sites. 2. Comparative validation dataset: chamber vs. direct sensor exposure, quantifying measurement benefit. 3. Performance characterization across two environments: accuracy, precision, drift, biofouling resilience. 4. First integrated model-measurement MRV validation dataset for open-system CDR. 5. Scientific paper draft and final report.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	This project resolves a concrete and overlooked challenge for marine CDR: the inability to make reliable, operational, and continuous carbonate system measurements in near-field environments. Moves near-field MRV from manual, campaign-based discrete sampling to continuous autonomous monitoring. Reduces uncertainty in net removal calculations by providing the near-field data that models need for validation. Enables MRV to scale without proportionally scaling field

	teams and using a cost-effective approach as foundation.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	18 months from contract execution. Month 2: Final design of both sampling system versions. Month 4: Manufacturing of 3 units complete. Month 5: Dry testing and wet testing (bench validation). Month 6: Deployment at both sites. Initial comparative data analysis. Month 9: First site inspection visits. Interim performance assessment. Month 12: Initial validation datasets complete. Pump vs. bladder comparison. Month 15: Recovery, inspection, biofouling and drift characterization. Month 18: Final report ready for submission and first draft of the paper ready for review.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$397,000 USD